

Recap

Passive monitoring techniques collect measurements by observing the behavior of a system while operating

Logging facilities collect measurements of the activities and events generated by applications and services

Performance monitors collect measurements of the hardware and software components

Performance profilers collect measurements of the applications

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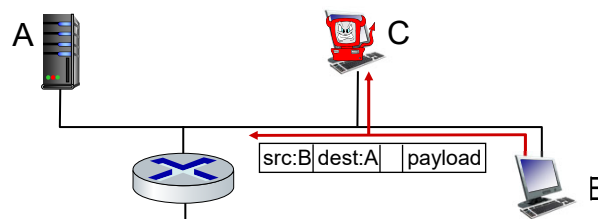
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Sniffing



Network sniffing means eavesdropping the traffic flowing on wired or wireless networks



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Sniffers

Sniffers are monitoring tools for network diagnostic and intrusion detection that collect measurements at a microscopic level by eavesdropping the traffic

They run on any device whose Network Interface Card is set to promiscuous mode (i.e., the NIC captures and stores all traffic flowing on the network)

The vantage point determines the portion of network being monitored

Measurements collected by sniffers refer to the attributes of the packets at each level of the protocol stack, e.g., timestamp, check-sum, length, MAC and IP addresses, source and destination port numbers, flags, time to live and **payloads** ... compliance with GDPR must be ensured

Depending on the objective of the monitoring project, measurements can be used either online or offline (i.e., live capture and offline analysis)

Serious issues concerning security and privacy (e.g., open WiFi)



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Sniffer implementation



Wireshark is the de facto standard used for network sniffing and analysis within enterprises and institutions

TShark is a terminal oriented version of Wireshark designed for capturing and displaying packets when an interactive user interface is not necessary or available

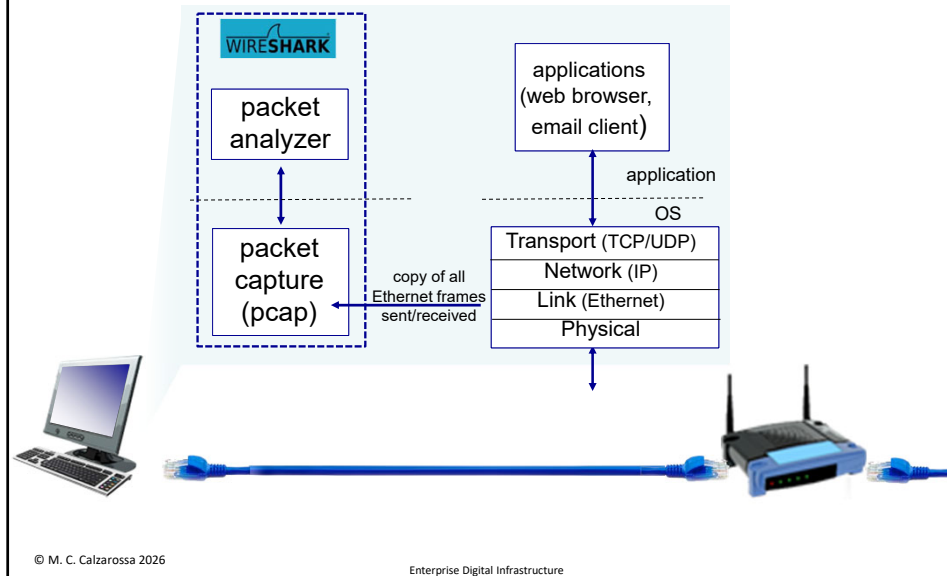
The **libpcap** is a portable C/C++ library developed within the **tcpdump** project and used to capture network traffic directly from NICs

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Wireshark



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Wireshark live capture

No.	Time	Source	Destination	Protocol	Length	Info
4010	9.449810	10.87.152.123	193.204.40.97	TCP	54	60875 → 3128 [ACK] Seq=589 Ack=2947 Win=131328 Len=0
4011	9.450003	10.87.152.123	193.204.40.97	TCP	54	60875 → 3128 [FIN, ACK] Seq=589 Ack=2947 Win=131328 Len=0
4012	9.450132	142.251.209.8	10.87.152.123	TCP	66	443 → 60879 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
4013	9.450214	10.87.152.123	142.251.209.8	TCP	54	60879 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=0
4014	9.451704	10.87.152.123	142.251.209.8	TCP	1466	60879 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=1412 [TCP PDU reassembled in 4015]
4015	9.451704	10.87.152.123	142.251.209.8	TLSv1.3	551	Client Hello (SNI=www.googletagmanager.com)
4016	9.452585	10.87.152.123	193.204.40.230	HTTP	515	GET /wp-includes/js/jquery/jquery-migrate.min.js?ver=1.4.1 HTTP/1.1
4017	9.453101	10.87.152.123	193.204.35.27	DNS	84	Standard query 0x2756 A www.googletagmanager.com
4018	9.454510	193.204.40.230	10.87.152.123	HTTP	238	HTTP/1.1 200 OK (text/css)

> Frame 4014: 1466 bytes on wire (11728 bits), 1466 bytes captured (11728 bits) on interface \Device\NPF_{6D0E9EDB-F001-4A51-ABE5-10C...}

> Ethernet II, Src: Microsoft_cb:7e:ae (bc:83:85:cb:7e:ae), Dst: Fortinet_09:00:1b (00:09:0f:09:00:1b)

> Internet Protocol Version 4, Src: 10.87.152.123, Dst: 142.251.209.8

> Transmission Control Protocol, Src Port: 60879, Dst Port: 443, Seq: 1, Ack: 1, Len: 1412

0020 d1 00 bd 41 01 bb
0030 02 00 ba a1 00 00
0040 03 36 23 55 a1 c5
0050 65 23 41 25 05 42
0060 d6 20 90 b3 08 a9
0070 c0 -1 ->4 -1c 4a -4a

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Transmission Control Protocol, Src Port: 60879, Dst Port: 443, Seq: 1, Ack: 1, Len: 1412

- Source Port: 60879
- Destination Port: 443
- [Stream index: 30]
- [Stream Packet Number: 4]
- > [Conversation completeness: Incomplete, DATA (15)]
- [TCP Segment Len: 1412]
- Sequence Number: 1 (relative sequence number)
- Sequence Number (raw): 521152425
- [Next Sequence Number: 1413 (relative sequence number)]
- Acknowledgment Number: 1 (relative ack number)
- Acknowledgment number (raw): 3726978039
- 0101 = Header Length: 20 bytes (5)
- ▼ Flags: 0x010 (ACK)
 - 000. = Reserved: Not set
 - ...0 = Accurate ECN: Not set
 - 0... = Congestion Window Reduced: Not set
 -0.. = ECN-Echo: Not set
 -0. = Urgent: Not set
 -1 = Acknowledgment: Set
 -0... = Push: Not set
 -0.. = Reset: Not set
 -0.. = Syn: Not set
 -0 = Fin: Not set

Internet Protocol Version 4, Src: 10.87.152.123, Dst: 142.251.209.8

- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 1452
- Identification: 0x8ae8 (35560)
- > 010. = Flags: 0x2, Don't fragment
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 128
- Protocol: TCP (6)
- Header Checksum: 0x678d [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 10.87.152.123
- Destination Address: 142.251.209.8
- [Stream index: 45]

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Wireshark capture: content

Frame 4767: 784 bytes on wire (6272 bits), 784 bytes captured (6272 bits) on interface \Device\NPF_{608E9EDB-F001-4A51-ABE5-10C908F3BF40}, id 0

- > Ethernet II, Src: TplinkTechno_95:d0:68 (98:48:27:95:d0:68), Dst: Microsoft_cb:7e:ae (bc:83:85:cb:7e:ae)
- > Internet Protocol Version 4, Src: 193.204.40.230, Dst: 192.168.1.103
- > Transmission Control Protocol, Src Port: 80, Dst Port: 50793, Seq: 201548, Ack: 6230, Len: 730
- ▼ Hypertext Transfer Protocol
 - > HTTP/1.1 200 OK\r\n
 - Date: Tue, 18 Mar 2025 19:55:43 GMT\r\n
 - Server: Apache/2.2.16 (Debian)\r\n
 - Last-Modified: Thu, 09 Feb 2017 08:27:40 GMT\r\n
 - Accept-Ranges: bytes\r\n
 - > Content-Length: 446\r\n
 - Keep-Alive: timeout=15, max=86\r\n
 - Connection: Keep-Alive\r\n
 - Content-Type: image/png\r\n
 - X-Pad: avoid browser bug\r\n
 - \r\n
 - [Request in frame: 4763]
 - [Time since request: 0.078100000 seconds]
 - [Request URI: /wp-content/themes/DiplunipIng/images/pagination.png]
 - [Full request URI: <http://webing.unipv.eu/wp-content/themes/DiplunipIng/images/pagination.png>]
 - File Data: 446 bytes
- ▼ Portable Network Graphics
 - PNG Signature: 89504e470d0a1a0a
 - > Image Header (IHDR)
 - > Textual data (tEXt)
 - > Image data chunk (IDAT)
 - > Image Trailer (IEND)

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Traffic analysis: expert information dialog

Wireshark · Expert Information · WiFi			
Severity	Summary	Group	Protocol
Error	Malformed Packet (Exception occurred)	Malformed	IEEE 802.11
Warning	ACKed segment that wasn't captured (common at capture start)	Sequence	TCP
Warning	Connection reset (RST)	Sequence	TCP
Warning	D-SACK Sequence	Sequence	TCP
Note	ACK to a TCP keep-alive segment	Sequence	TCP
Note	TCP keep-alive segment	Sequence	TCP
Note	Partial Acknowledgement of a segment	Sequence	TCP
Note	Time To Live	Sequence	IPv4
Note	This frame is a (suspected) spurious retransmission	Sequence	TCP
Note	This frame undergoes the connection closing	Sequence	TCP
Note	This frame initiates the connection closing	Sequence	TCP
Note	The acknowledgment number field is nonzero while the ACK flag is not set	Protocol	TCP
Note	Duplicate ACK	Sequence	TCP
Note	This frame is a (suspected) retransmission	Sequence	TCP
Chat	Connection finish (FIN)	Sequence	TCP
Chat	This legacy_version field MUST be ignored. The supported_versions exte...	Deprecated	TLS
Chat	Connection establish acknowledge (SYN+ACK)	Sequence	TCP
Chat	Connection establish request (SYN)	Sequence	TCP

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Traffic analysis: endpoints

Wireshark · Endpoints · WiFi

Endpoint Settings

☐ Name resolution

☐ Limit to display filter

Copy

Map

Protocol

☒ Ethernet

☐ FC

☐ FDDI

☐ IEEE 802.11

☐ IEEE 802.15.4

☒ IPv4

☒ IPv6

Filter list for specific type

Ethernet · 343

IPv4 · 83

IPv6 · 8

TCP · 61

UDP · 113

Address	Port	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
4.207.247.137	443	8	883 bytes	3	524 bytes	5	359 bytes
10.87.148.122	7680	5	330 bytes	0	0 bytes	5	330 bytes
10.87.152.123	49426	2	121 bytes	1	55 bytes	1	66 bytes
10.87.152.123	56593	3	381 bytes	2	152 bytes	1	229 bytes
10.87.152.123	56594	3	381 bytes	2	152 bytes	1	229 bytes
10.87.152.123	56648	2	121 bytes	1	55 bytes	1	66 bytes
10.87.152.123	60606	33	3 kB	13	1 kB	20	2 kB
10.87.152.123	60610	12	2 kB	5	825 bytes	7	706 bytes
10.87.152.123	60644	4	282 bytes	2	108 bytes	2	174 bytes
10.87.152.123	60684	14	1 kB	6	634 bytes	8	693 bytes
10.87.152.123	60727	14	1 kB	6	630 bytes	8	689 bytes
10.87.152.123	60756	19	5 kB	8	4 kB	11	2 kB
10.87.152.123	60758	11	6 kB	4	3 kB	7	4 kB
10.87.152.123	60822	2,319	2 MB	465	188 kB	1,854	2 MB
10.87.152.123	60854	5	330 bytes	5	330 bytes	0	0 bytes
10.87.152.123	60855	5	330 bytes	5	330 bytes	0	0 bytes

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Traffic analysis: Packet lengths

Wireshark · Packet Lengths · WiFi								
Topic / Item	Count	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
▼ Packet Lengths	7348	549.57	42	1514	0.3040	100%	5.0500	6.181
0-19	0	-	-	-	0.0000	0.00%	-	-
20-39	0	-	-	-	0.0000	0.00%	-	-
40-79	4142	57.02	42	79	0.1714	56.37%	1.0200	11.365
80-159	327	98.84	80	157	0.0135	4.45%	0.2900	5.305
160-319	154	250.68	161	317	0.0064	2.10%	0.1300	7.346
320-639	205	474.06	324	638	0.0085	2.79%	0.2300	9.434
640-1279	162	940.54	642	1265	0.0067	2.20%	0.1600	6.219
1280-2559	2358	1476.50	1285	1514	0.0976	32.09%	4.3300	6.181
2560-5119	0	-	-	-	0.0000	0.00%	-	-
5120 and greater	0	-	-	-	0.0000	0.00%	-	-

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Microsoft Packet Monitor

Packet Monitor (**pktmon**) is a built-in sniffer for Microsoft Windows environments (e.g., MS Azure) aimed at network diagnostics

It is very useful in virtualized scenarios (e.g., Software Defined Networks) because it provides visibility within the entire network stack

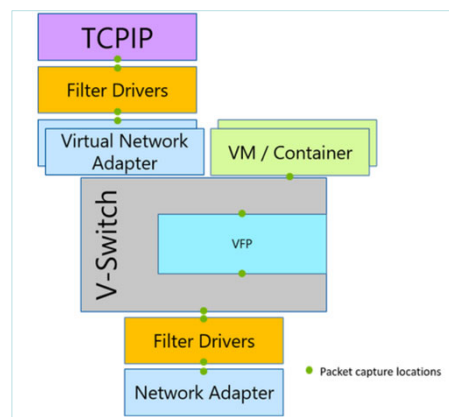
It can be used for packet capture, drop detection, filtering and counting

Data is saved into Event Trace Log (ETL) files that can be converted into **pcap** format and analyzed using Wireshark)

No GUI

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pktmon

```
PS C:\WINDOWS\system32> pktmon
pktmon <command> [OPTIONS | help]
Advanced packet capture and event collection.
```

```
Commands
  filter      Manage packet filters.
  list        List packet processing components.

  start       Start packet capture and event collection.
  stop        Stop data collection.
  status      Query current status.
  unload      Unload PktMon driver.
```

```
counters     Display current packet counters.
reset        Reset packet counters to zero.
```

```
etl2txt      Convert log file to text format.
etl2pcap     Convert log file to pcap format.
hex2pkt      Decode packet in hex to pcap format.
```

```
help         Show help text for pktmon s
Example: pktmon s
```

Marvell AVASTAR Wireless-AC Network Controller									
Id Name		Counter	Direction	Packets	Bytes	Direction	Packets	Bytes	

12 Marvell AVASTAR Wireles...		Drops	Rx	1	86	Tx	29	1,742	
		Upper	Rx	47,037	43,431,141	Tx	35,486	6,016,562	

40 WFP Native Filter		Lower	Rx	47,037	43,431,141	Tx	35,486	6,016,562	
		Upper	Rx	47,037	43,431,141	Tx	35,486	6,016,562	
39 Npcap Packet Driver (NP...		Lower	Rx	47,037	43,431,141	Tx	35,486	6,016,562	
		Upper	Rx	47,037	43,431,141	Tx	35,486	6,016,562	
38 Virtual WiFi Filter Driver		Lower	Rx	47,037	43,431,141	Tx	35,486	6,016,562	
		Upper	Rx	47,037	43,431,141	Tx	35,486	6,016,562	
37 Native WiFi Filter Driver		Lower	Rx	47,037	43,431,141	Tx	35,486	6,016,562	
		Upper	Rx	47,031	42,491,191	Tx	35,480	5,377,070	
36 Npcap Packet Driver (NP...		Lower	Rx	47,031	42,491,191	Tx	35,480	5,377,070	
		Upper	Rx	47,031	42,491,191	Tx	35,480	5,377,070	
35 QoS Packet Scheduler		Lower	Rx	0	0	Tx	35,480	5,377,070	
		Upper	Rx	0	0	Tx	35,480	5,377,070	
34 WFP 802.3 Filter		Lower	Rx	47,031	42,491,191	Tx	35,480	5,377,070	
		Upper	Rx	47,031	42,491,191	Tx	35,480	5,377,070	

73 TCP/IP		Lower	Rx	47,030	42,491,105	Tx	35,509	5,378,812	
71 LLTDIO		Lower	Rx	47,031	42,491,191	Tx	0	0	
69 NDISUIO		Lower	Rx	47,031	42,491,191	Tx	0	0	

86 HTTP (192.168.1.104)		Lower	Rx	2	959	Tx	1	2,351	

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Measurement platforms

Measurement platforms refer to technological infrastructures used to collect performance measurements

Probes, i.e., measurement points, available at strategic locations within access and backbone networks and behind residential gateways

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Measurement Lab (M-Lab) is a non-profit large project supported by organizations, educational institutions and private companies dedicated to

- Provide an open, verifiable measurement platform for global network performance
- Host the largest open Internet performance dataset on the planet
- Create visualizations and tools to help people make sense of Internet performance

M-Lab's mission is to advance Internet research by empowering consumers with useful information about their Internet performance

From measurements, researchers, regulators, and general public can understand how the Internet is working and how to maintain and improve it

Data collected by M-Lab global measurement platform are made openly available (including the IP addresses that users use for the test)

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Measurement Lab

The platform provides open-source active tools used to collect measurements



NDT (Network Diagnostic Tool)
Tests your connection speed, and provides a sophisticated diagnosis of problems limiting speed.



Neutbot DASH
DASH is designed to measure the quality of tested networks by emulating a video streaming player.



Reverse Traceroute
Measures the network path back to a user from selected network endpoints.



WeHe
WeHe uses your device to exchange Internet traffic recorded from real, popular apps like YouTube and Spotify, and attempts to tell you whether your ISP is giving different performance to an app's network traffic.



MSAK
MSAK (Measurement Swiss-Army Knife) implements a configurable Websocket-based throughput measurement protocol, and a UDP-based latency measurement protocol.

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M-Lab infrastructure

The M-Lab infrastructure is composed by widely distributed servers

Each physical “site” includes of 3-4 servers and one switch, connected directly to an upstream provider

Virtual sites include one or more virtual servers proportionate to demand

Multiple sites are located in large metro area to obtain diversity in routes



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[RIPE Atlas](#) is a measurement platform deployed by the RIPE Network Coordination Centre (NCC) for collecting Internet measurements

It employs a global network of probes (i.e., small, USB-powered hardware devices connected to an Ethernet port on a router) that measure Internet connectivity and reachability



It also employs anchors, i.e., probes designed for highly-available networks and physical environments to operate as stable measurement devices

Probes and anchors perform active measurements by generating traffic, e.g., ping, traceroute, DNS and NTP queries, TLS traffic

The collected data is aggregated and made publicly available

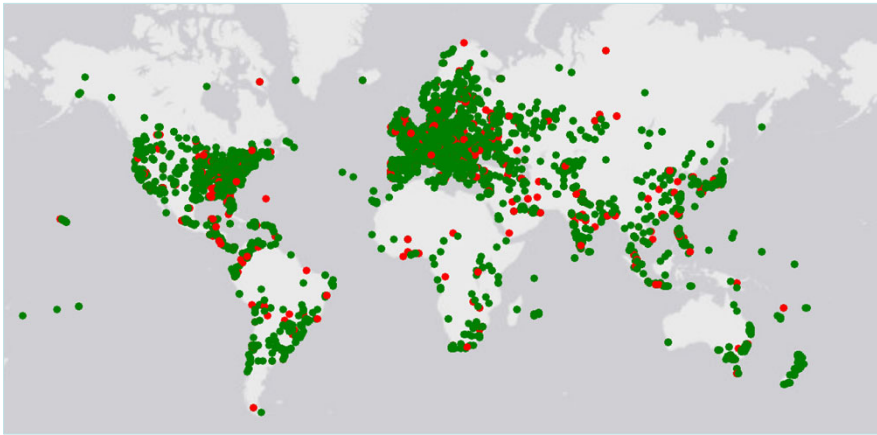
Users hosting a probe can use the entire RIPE Atlas network for customised measurements

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RIPE Coverage

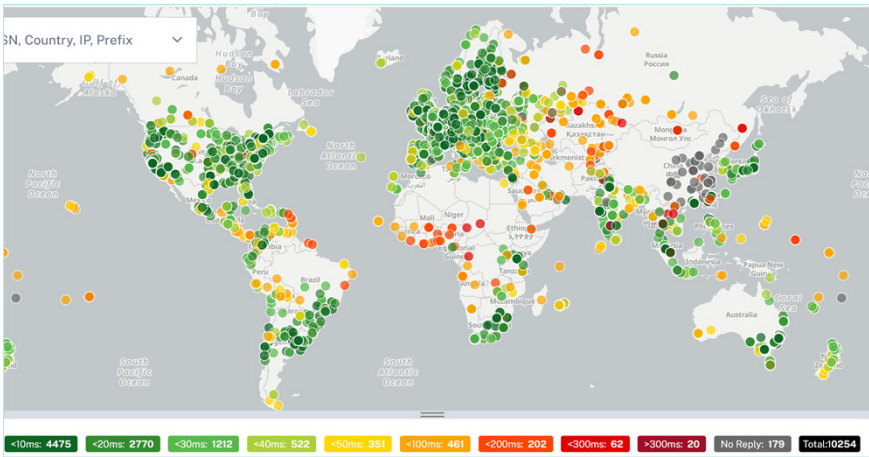


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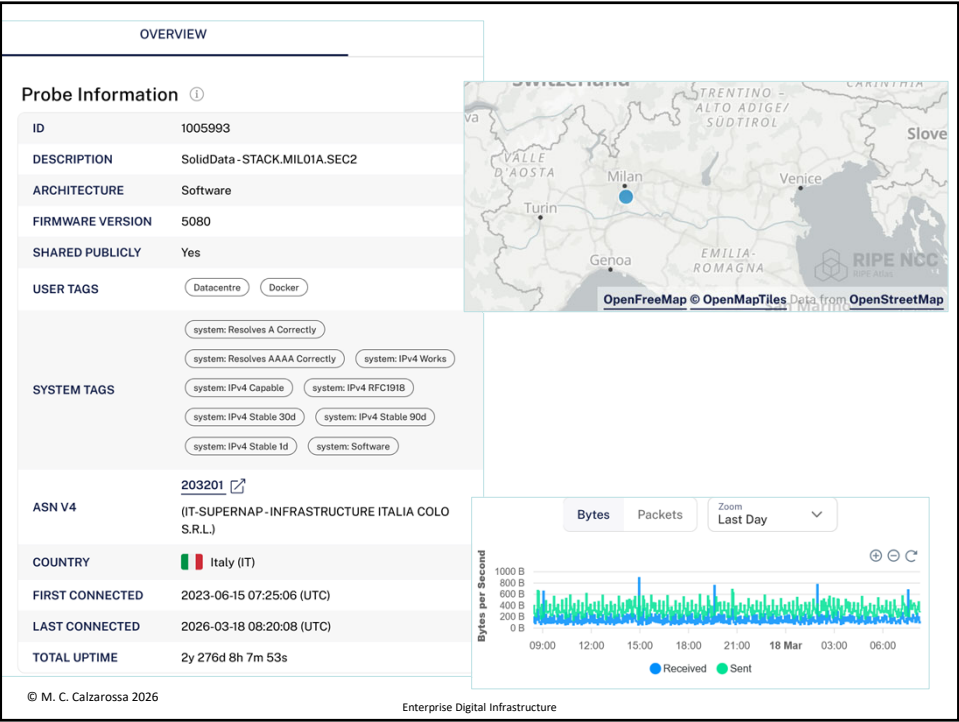
RTTs towards Google



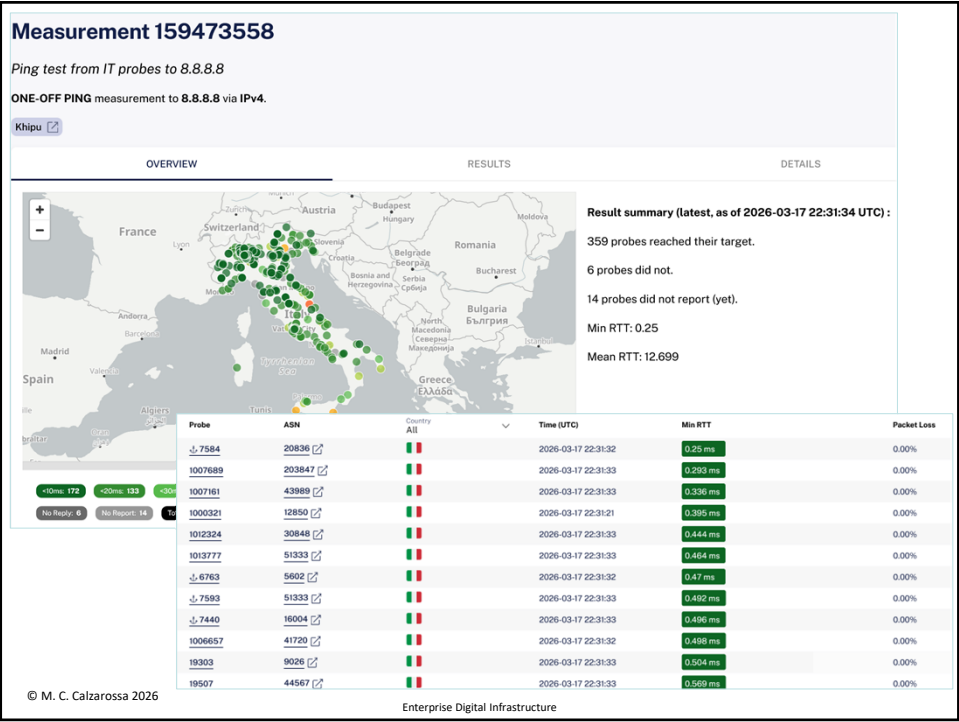
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